PATHFINDER

AN INDOOR NAVIGATION WEB APPLICATION FOR THE PHYSICAL SCIENCES BUILDING IN THE UNIVERSITY OF THE PHILIPPINES LOS BAÑOS

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ABSTRACT

This study developed and evaluated a web-based indoor navigation application for the Physical Sciences Building at the University of the Philippines Los Banos, designed as a scalable and software-driven alternative to hardware-based positioning and static physical floor plans. The application provides a navigation interface that guides users between selected origin and destination points, and a layout builder that allows building administrators to create and publish digital map layouts for display within the application.

OBJECTIVES

- Integrate point to point directions to aid users in finding the exact locations of classrooms, offices, and facilities within the Physical Sciences Building.
- Enable building administrators to generate a building's maps based on floor plans, allowing for updates and customization of the mapping content.
- Assess the usability of the system through user testing to evaluate navigation accuracy, interface intuitiveness, and overall user satisfaction.

METHODOLOGY

A 2D layout of the building was provided, separated into three files, with each file mapping out an entire floor. A physical visit inside the building was conducted to cross-examine the accuracy of the floor plan against the actual rooms

The application was developed to enable users to search for specific destinations within the building by providing an origin and a destination. It also includes a layout builder that enables administrators to create and edit an indoor map to be seen by the users.

User testing was conducted with two participant groups: regular users and building administrators. Building administrators tested the map layout builder by performing editing functions such as adding, modifying, saving, and publishing map elements, then verified that changes were reflected in the navigation interface. Regular users tested navigation accuracy to evaluate whether the application correctly guided them from a selected origin to a destination within the building.

A total of ten (10) responses were collected, with four coming from building administrators from the University Planning and Maintenance Office (UPMO) and six from students studying at the University of the Philippines Los Baños.

RESULT

- Qualitatively, all six student respondents confirmed that the application successfully led them to their intended destination and described the routes as accurate or mostly accurate.
- The four building administrators who tested the layout builder feature described the tool as generally intuitive and responsive, useful for early-stage planning of floor layouts.

RESPONDENT SCORES

TABLE I: Raw Scores per Respondent

Respondent	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
R1	4	1	5	1	5	3	5	5	5	5
R2	3	1	5	1	4	3	4	1	5	1
R3	4	2	4	2	5	3	5	4	4	4
R4	5	1	5	1	5	1	5	1	5	2
R5	5	1	5	1	5	1	5	1	5	1
R6	3	1	4	3	4	1	4	2	4	5

TABLE II: Total Scores by Respondent and Score Rating

Respondent	SUS Score	Rating
R1	72.5	Good
R2	85.0	Excellent
R3	67.5	OK
R4	97.5	Excellent
R5	100	Excellent
R6	67.5	OK

CONCLUSION

The two testing groups suggested that the application performed strongly on its core function, indoor navigation, while leaving room for refinement in the layout builder. Users rated the navigation experience highly, with a mean score falling within the ”Excellent” range. Building administrators, on the other hand, found the layout builder and sandbox mode useful for building layout planning, helping communicate layout proposals to stakeholders and clients.